Highlights

Mango-Mamba and VN-MangoLeaf: A Lightweight Mamba Model and New Dataset for Mango Leaf Disease Classification

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- MangoMamba achieves $\leq$ 1.52 ms GPU inference with competitive accuracy.
- Hybrid Mamba-Attention model offers linear complexity for disease detection.
- VN-MangoLeaf dataset introduced: 7000 images from Vietnam with Red Rust.
- Three-phase curriculum learning enables cross-geographical knowledge transfer.
- Lightweight $\leq$ 25 MB model enables 100-300 ms smartphone-based detection.

Mango-Mamba and VN-MangoLeaf: A Lightweight Mamba Model and New Dataset for Mango Leaf Disease Classification

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ARTICLE INFO

Keywords:
Deep Learning
Mango Leaf Classification
Mamba Models
Image processing
Smart agriculture

ABSTRACT

Mango leaf disease represents a significant threat to fruit quality and yield, necessitating highly accurate, real-time detection systems. However, existing Deep Learning approaches, particularly Transformer-based models, often suffer from prohibitive computational complexity (quadratic scaling), limiting their deployment on resource-constrained edge devices. To address this challenge, this study introduces MangoMamba, a novel lightweight hybrid architecture specifically optimized for mobile deployment. The proposed model integrates Multi-Scale Mamba Mixers with Large-Kernel Attention mechanisms within a hierarchical four-stage framework, enabling linear computational complexity while preserving global receptive fields. Experimental evaluations were conducted on the MangoLeafBD dataset and the newly curated VN-MangoLeaf dataset, which comprises 7,000 images of Vietnamese mango varieties. Results demonstrate that MangoMamba achieves competitive classification accuracies of 99.75% and 98.71% on the respective datasets. Crucially, the model exhibits exceptional efficiency with only 5.8 million parameters and an inference latency of 1.46 ms per image on T4 GPU, approximately 80 times faster than recent ViX-MangoEFormer architectures. Furthermore, the practical feasibility of the proposed approach is validated through a functional Android application capable of offline inference (100-300 ms latency) on standard smartphones. These findings confirm that MangoMamba establishes a new competitive trade-off between accuracy and efficiency for smart agriculture applications.

1. Introduction

Mango (*Mangifera indica* L.) cultivation is a vital component of the global agricultural economy, yet it faces substantial production risks due to foliar pathogens. Diseases such as Anthracnose, Powdery Mildew, and Red Rust can cause yield losses exceeding 20-30% if not managed effectively [1]. Traditional disease management relies on manual visual inspection by pathologists, a process that is labor-intensive, subjective, and difficult to scale in large orchards or resource-limited regions [9]. Consequently, the development of automated, accurate, and real-time disease detection systems has become a priority in smart agriculture.

Recent advancements in deep learning have significantly improved automated plant disease diagnosis. Convolutional Neural Networks (CNNs) [14, 8] and Transformer-based architectures [6, 2] have achieved high classification accuracy on benchmark datasets. However, the practical deployment of these models on edge devices (e.g., smartphones, embedded sensors) is hindered by computational constraints. While Vision Transformers (ViTs) offer superior global context modeling compared to CNNs, they suffer from a quadratic computational complexity ($O(N^2)$) with respect to image resolution due to the self-attention mechanism [7]. This results in high memory consumption and latency, rendering them unsuitable for real-time applications in rural environments where internet connectivity for cloud processing is often unreliable [9].

Additionally, recent studies have demonstrated the potential of advanced hybrid architectures in addressing performance and computational challenges. For instance, Qin et al. [11] introduced CropCapsNet, which integrates an enhanced capsule network with Squeeze-and-Excitation Inception modules. This approach effectively captures multi-scale features for crop disease classification, achieving remarkable accuracy by overcoming the spatial information loss inherent in traditional CNNs. Similarly, Truong et al. [17] proposed SC-MambaFew, utilizing the Mamba architecture coupled with selective spatial-channel attention for bearing fault diagnosis. Their work highlights the capability of Mamba-based hybrids to model long-range dependencies efficiently, particularly in few-shot learning

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scenarios. However, while these pioneering works validate the efficacy of hybrid models, they do not explicitly target the optimization of Mamba-Attention mechanisms for the subtle, fine-grained visual characteristics of mango leaf diseases on strictly resource-constrained edge hardware. Furthermore, the issue of cross-domain generalization against geographical bias- a critical requirement in global agriculture-remains largely unaddressed in these architectural studies. This research builds upon these advancements to bridge that specific gap.

To address the limitations of quadratic complexity while ensuring high diagnostic precision, State Space Models (SSMs), particularly the Mamba architecture [3], have emerged as a promising alternative. Mamba introduces a selective scan mechanism that allows for modeling long-range dependencies with linear computational complexity ($O(N)$). Recent adaptations of Mamba for computer vision, such as MambaVision [5] and Vision Mamba [22], have demonstrated that SSMs can rival Transformers in performance while significantly reducing inference costs. Despite these advances, the specific optimization of hybrid Mamba-Attention architectures for lightweight agricultural disease classification remains underexplored. Existing lightweight models often sacrifice accuracy for speed or fail to capture the multi-scale features required to distinguish between morphologically similar mango leaf diseases [21].

Furthermore, a critical gap exists in the geographical diversity of available mango leaf disease datasets. The majority of public datasets, including MangoLeafBD [1] and MLD24 [15], originate primarily from Bangladesh. This geographical bias limits the generalization capability of models when applied to regions with different mango cultivars and disease prevalence, such as Vietnam, where distinct pathogens like Red Rust are more common [10].

To bridge these gaps, this study proposes **MangoMamba**, a lightweight architecture specifically engineered for efficient mango leaf disease classification on edge devices. The architecture integrates Multi-Scale Mamba Mixers for linear-complexity global context modeling with Large-Kernel Attention (LKA) mechanisms for local feature refinement. Additionally, the **VN-MangoLeaf** dataset is introduced to address regional data scarcity, complemented by a curriculum learning strategy to enhance cross-domain generalization.

The primary contributions of this research are summarized as follows:

1. A specialized hybrid architecture, MangoMamba, is developed, combining the efficiency of Selective State Space Models with the spatial precision of Large-Kernel Attention. This design achieves linear computational complexity, significantly reducing inference latency compared to Transformer-based counterparts.
2. The VN-MangoLeaf dataset is introduced, comprising 7,000 images collected from Vietnamese orchards. This dataset mitigates geographical bias by including regionally specific classes such as Red Rust, enabling more robust model evaluation.
3. A three-phase curriculum learning methodology is implemented to facilitate effective knowledge transfer between datasets with varying class distributions, enhancing the model's robustness against morphological variations.
4. Extensive experimental validation demonstrates that the proposed model achieves competitive accuracy (99.75% on MangoLeafBD and 98.14% on VN-MangoLeaf) while maintaining an inference time of approximately 1.5 ms per image, verifying its suitability for real-time mobile deployment.

2. Materials and Methods

Datasets overview

Existing public datasets for mango leaf disease detection, such as MangoLeafBD [1] and MLD24 [15], have provided valuable benchmarks for the research community. However, these datasets originate predominantly from Bangladesh, which limits the generalization capability of models when applied to other geographical regions with different environmental conditions and pathogen distributions. To address this geographical bias and the lack of diverse regional data, this study introduces the **VN-MangoLeaf** dataset, collected extensively across southern Vietnam.

A critical distinction of the VN-MangoLeaf dataset is the specific disease distribution observed in Vietnam compared to Bangladesh (Figure 1). While diseases such as 'Cutting Weevil' and 'Bacterial Canker' are common in previous datasets, they are infrequently observed in Vietnamese mango orchards. Conversely, 'Red Rust', caused by the algal plant pathogen *Cephaleuros virescens* [10], is highly prevalent in this region but underrepresented in existing benchmarks. Red Rust manifests as orange-brown, rust-like spots, typically on the upper leaf surface [10], as illustrated in Figure 2.

The dataset comprises seven distinct classes: Healthy, Anthracnose, Gall Midge, Powdery Mildew, Sooty Mold, Dieback, and Red Rust. Images were acquired using a Xiaomi Redmi Note 11S smartphone camera. To ensure high

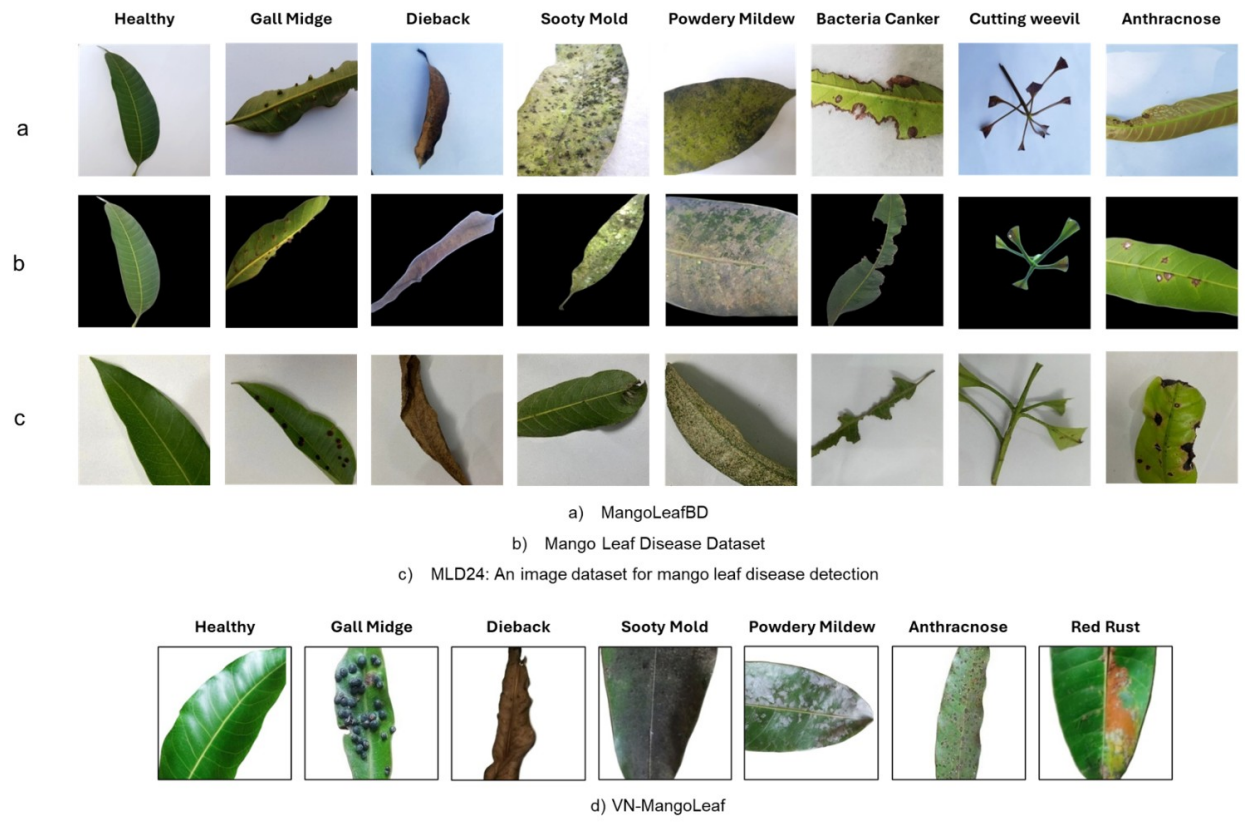


Figure 1: Comparison of disease classes: 8 classes from existing Bangladesh-based datasets versus 7 classes from the proposed VN-MangoLeaf dataset.

fidelity, images were captured at a native resolution of 3000×4000 pixels under natural lighting conditions between 7:00 AM and 4:00 PM, capturing variations in sunlight intensity and shadowing. The collection sites covered diverse locations in Southern Vietnam, including Binh Thanh District, District 12, District 9 (Ho Chi Minh City), Cao Lanh City (Dong Thap Province), and Giong Trom District (Ben Tre), as described in Figure 3.

To ensure label accuracy and dataset consistency, a two-stage verification process was implemented. Initial labeling was performed by the research team, followed by cross-verification by experienced mango orchard owners, who acted as domain experts. This process resulted in the removal of approximately 2,000 images identified as ambiguous, mislabeled, or of poor quality. The final curated dataset comprises 7,000 images (post-balancing), with at least 700 verified original images per class prior to augmentation. A comparison between VN-MangoLeaf and existing datasets is presented in Table 1.

2.1. Data augmentation

To facilitate the proposed Curriculum Learning strategy, this study utilizes two distinct categories of data serving complementary roles: a large-scale **Combined Dataset** serving as the source domain for general feature learning, and the proprietary **VN-MangoLeaf** dataset serving as the target domain for regional adaptation. Three publicly available datasets originating from Bangladesh: MangoLeafBD [1], MLD24 [15], and the Mendeley Mango Leaf Disease Dataset [12] were consolidated. This merged corpus, referred to as the *Combined Dataset*, provides a diverse range of 8 disease classes but suffers from significant class imbalance (Figure 4). It's primarily used in the early training phases to expose the model to a wide variety of visual symptoms.

In contrast, the **VN-MangoLeaf** dataset (described in Section 2) represents the specific environmental conditions and pathogen distributions of Vietnam. It is carefully verified by experts, contains the unique 'Red Rust' class, and excludes classes not prevalent in the region (Cutting Weevil and Bacterial Canker). This dataset is reserved for the



Figure 2: Representative samples of Mango leaves infected with Red Rust (*Cephaleuros virescens*) from the VN-MangoLeaf dataset.



Figure 3: The mango orchards locations of VN-MangoLeaf

final fine-tuning phase and performance benchmarking to validate the model's real-world applicability in the target geographical region.

As illustrated in Figure 4, the initial distribution of the three Bangladesh mango leaf datasets exhibited significant imbalance, with classes such as Bacterial Canker being overrepresented compared to Powdery Mildew. Such disparities can lead to model overfitting on the majority classes.

Table 1
Comparison between mango leaf disease datasets from Bangladesh and our proposed dataset from Vietnam

Dataset	Data format	Resolution	Image amount	Type of disease	Class	Image per class	Locations of images
VN-MangoLeaf	JPG	240x320	7000 (4900 original + 2100 augmented)	Healthy; Anthracnose; Powdery Mildew; Dieback; Sooty Mold; Gall Midge; Red Rust	7	1000	Giong Trom (Ben Tre); Cao Lanh (Dong Thap); District 12 (TPHCM); Binh Thanh (TPHCM); District 9 (TPHCM)
MangoLeafBD	JPG	240x320	4000 (1800 original + 2200 augmented)	Healthy; Anthracnose; Powdery Mildew; Dieback; Sooty Mold; Gall Midge; Bacterial Canker; Cutting Weevil	8	500	Sher-e-Bangla Agricultural University orchard; Jahangir Nagar University orchard; Udaypur village orchard; Itakhola village orchard (Bangladesh)
MLD24	JPG	240x240	6400 (all original)	Healthy; Anthracnose; Powdery Mildew; Dieback; Sooty Mold; Gall Midge; Bacterial Canker; Cutting Weevil	8	800	Kushtia; Dhaka (Bangladesh)

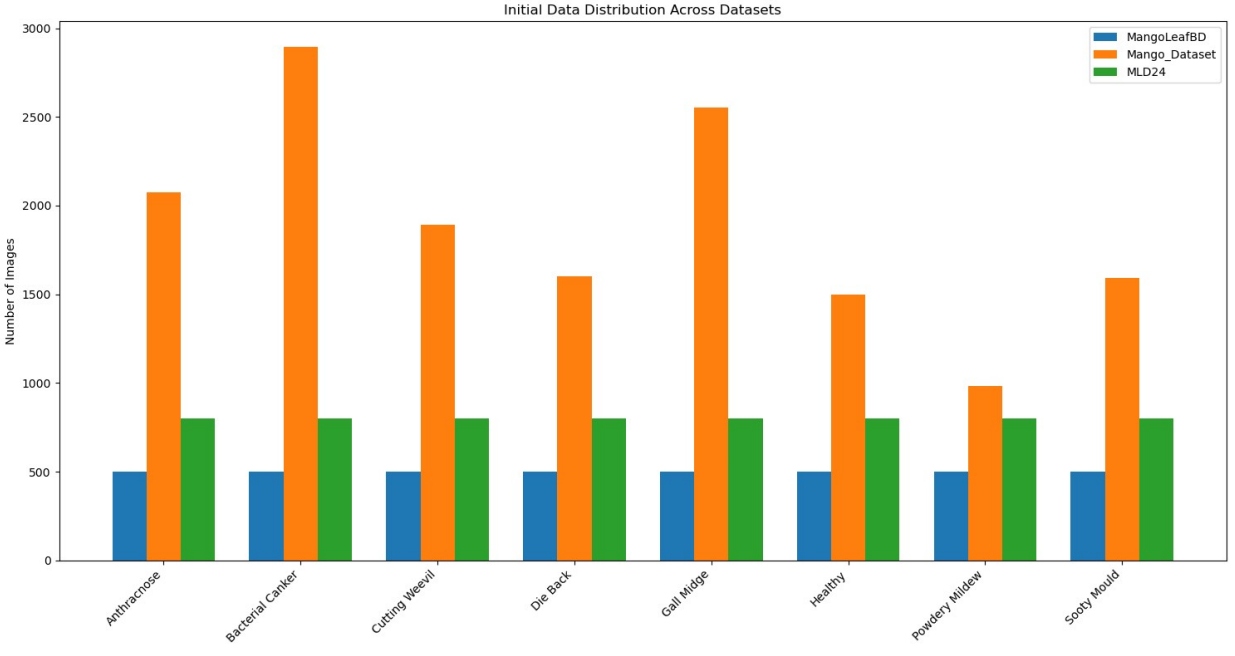


Figure 4: The initial class distribution across the component datasets before balancing, highlighting significant imbalances.

To mitigate this, augmentation strategies adapted from Noman et al. [9] were employed. Geometric transformations (rotation, flipping) and photometric adjustments (brightness, scaling) were applied specifically to underrepresented classes to achieve a target balance. Table 2 details the augmentation parameters and preprocessing steps.

Following these procedures, the Combined Dataset achieved a balanced distribution with approximately 5,000 images per class, totaling 40,000 images. Figure 5 depicts the class distribution post-augmentation.

Table 2
Data augmentation parameters and preprocessing specifications.

Processing Step	Description & Parameters
1. Augmentation	Applied stochastically to achieve target counts
Rotation	Range: $[-45°, 45°]$
Flipping	Vertical and Horizontal
Brightness	Factor range: $[0.8, 1.2]$
Scaling	Factor range: $[0.5, 1.5]$
2. Resizing	Uniform resizing to 224×224 pixels (JPG format)
3. Normalization	Min-Max scaling to the range $[0, 1]$ for stable convergence.

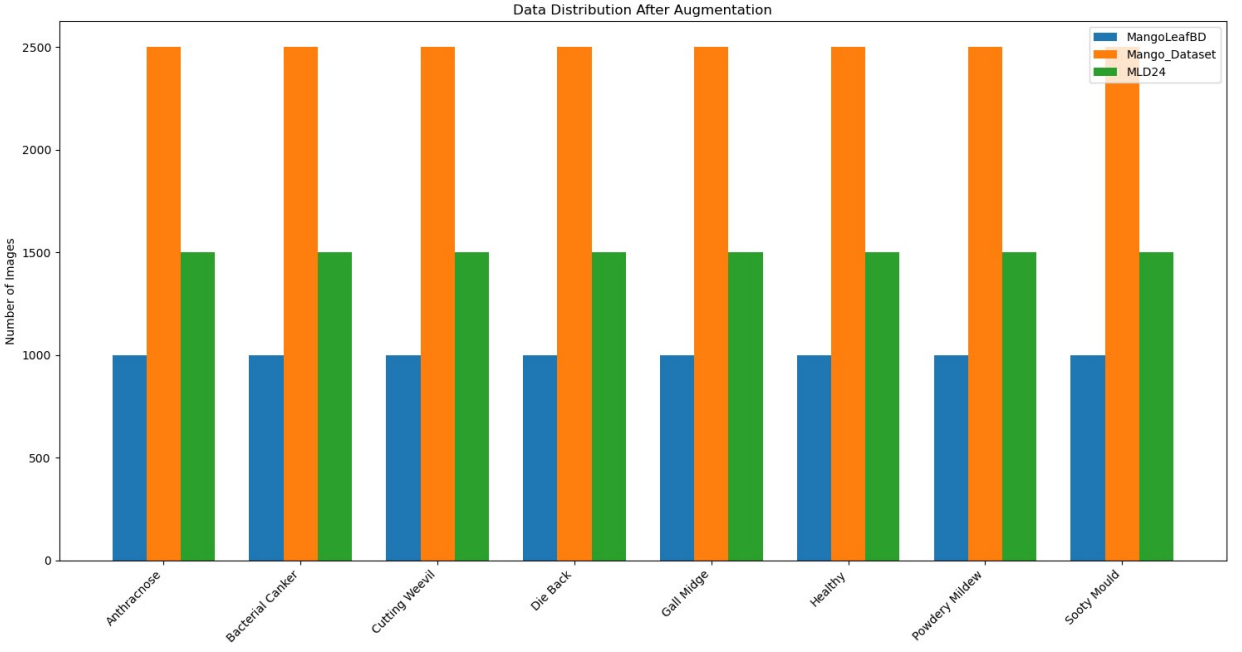


Figure 5: Balanced class distribution of the Combined Dataset following the application of augmentation strategies.

To ensure consistent feature scaling and accelerate gradient descent convergence, all images underwent Min-Max normalization. Each pixel value x was scaled to x' using Equation 1:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

where $x_{\max}$ and $x_{\min}$ represent the maximum (255) and minimum (0) pixel intensities, respectively.

To strictly evaluate generalization performance and prevent data leakage, the datasets were partitioned using a stratified random split with a ratio of 80:10:10 for training, validation, and testing, respectively. The specific distribution of images across these subsets for the MangoLeafBD, Combined, and VN-MangoLeaf datasets is summarized in Table 3.

2.2. MangoMamba model's architecture

The proposed MangoMamba architecture is a lightweight, hierarchical vision backbone specifically engineered for mango leaf disease recognition on edge devices. It prioritizes a compact model size and rapid inference capabilities

Table 3
Distribution of images across Training, Validation, and Test sets for each experimental dataset.

Dataset	Train (80%)	Validation (10%)	Test (10%)
MangoLeafBD	3,200	400	400
Combined Dataset	32,000	4,000	4,000
VN-MangoLeaf	5,600	700	700

while maintaining high classification accuracy. The architecture adopts a four-stage structure with progressively decreasing spatial resolutions ($H/4 \rightarrow H/32$) and increasing channel dimensions ($C \rightarrow 8C$). To maintain linear computational complexity, each stage processes features through non-overlapping windows.

Unlike standard hybrid architectures that combine State Space Models (SSM) with computationally intensive Transformer blocks, MangoMamba introduces a hybrid integration of Mamba Mixers and Large-Kernel Attention (LKA). This design strategically utilizes the global receptive field of SSMs for sequence modeling and the spatial adaptability of LKA for capturing local disease textures, effectively eliminating the quadratic complexity associated with traditional Self-Attention.

The model integrates several advanced components, selected based on extensive ablation studies to maximize the trade-off between efficiency and performance:

- **Global Response Normalization (GRN):** Utilized to enhance the contrast and selectivity of channels, thereby preventing feature collapse and stabilizing training dynamics. This component, adapted from ConvNeXt V2 [20], ensures that feature diversity is maintained across the deep layers of the network.
- **Efficient Channel Attention (ECA):** Implemented to provide channel-wise feature enhancement with negligible parameter overhead [19]. By emphasizing informative channels, ECA allows the model to focus on distinct disease characteristics amidst complex leaf backgrounds.
- **Windowed Processing:** Inspired by the hierarchical design of the Swin Transformer [7], feature maps are partitioned into local windows. This approach facilitates efficient local attention computation and hierarchical feature representation, eliminating the heavy computational burden associated with global attention.
- **Multi-Scale Mamba Mixers:** Inspired by the MambaVision architecture [5], this component is adapted to model multi-resolution features. It preserves the linear scaling superiority of SSMs [3], enabling the model to effectively capture long-range dependencies across the leaf surface.
- **Large-Kernel Attention (LKA):** Incorporated to replace the Transformer component found in typical hybrid models. LKA decomposes large kernel convolutions to capture spatial correlations and long-range dependencies with lower computational cost and parameter count compared to standard self-attention mechanisms [4].
- **SwiGLU Activation:** Employed within the Multi-Layer Perceptron (MLP) blocks, SwiGLU offers enhanced representational capacity compared to standard GELU or ReLU activations [16]. It operates as a gated linear unit with a sigmoid-activated component, improving the model's ability to learn complex non-linear disease patterns.

As visualized in Figure 6, the input processing commences with a lightweight patch embedding module. This module efficiently transforms raw images into patch-level representations using two consecutive stride-2 convolutions, reducing the spatial dimension to a quarter of the input resolution while projecting to the base channel dimension.

To ensure robust feature initialization, this embedding is immediately refined by the ECA and GRN mechanisms. The ECA module generates channel-wise attention weights by applying a parameter-efficient 1D convolution over global average pooling features, followed by a sigmoid activation. These weights are then element-wise multiplied with the input features to recalibrate channel importance. Subsequently, the GRN mechanism stabilizes feature responses by normalizing the spatial L2-norm across height and width dimensions, followed by a learnable affine transformation. This process emphasizes spatially important regions and suppresses redundant information.

The core of the architecture employs a hierarchical four-stage design utilizing windowed processing. In each stage, feature maps are partitioned into non-overlapping windows of size $S \times S$. This partitioning transforms the input tensor

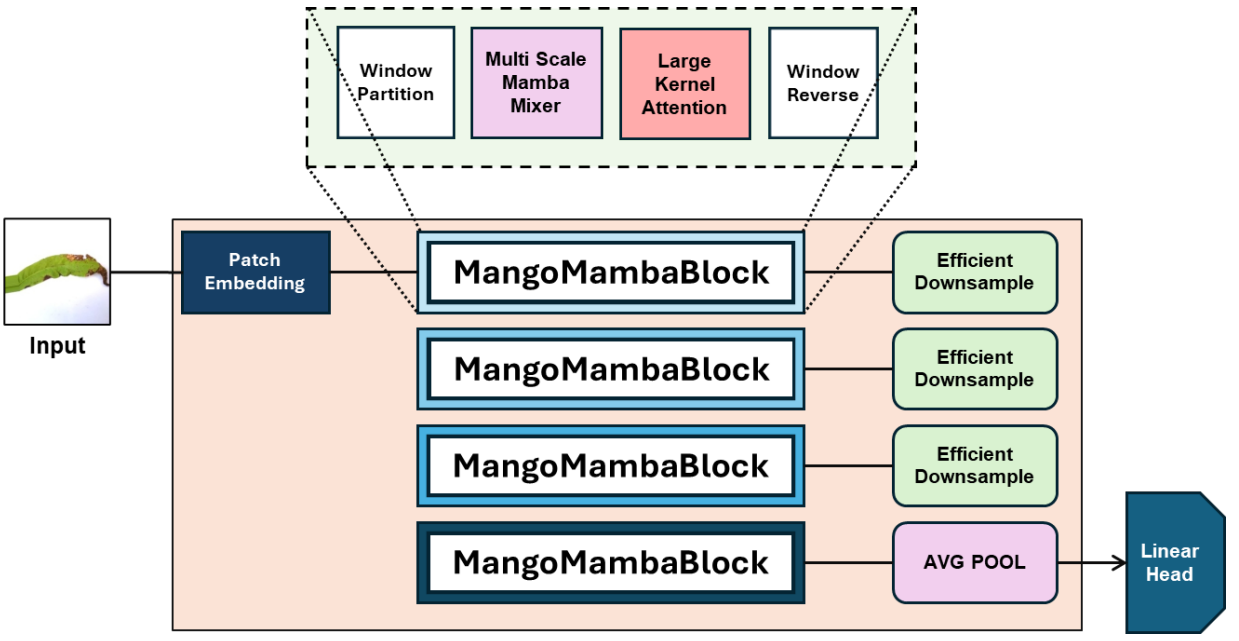


Figure 6: The architectural overview of the proposed MangoMamba model

into a windowed sequence, allowing the model to perform computations on smaller, fixed-size token sets rather than the entire image at once. This strategy significantly reduces memory usage and computational complexity.

Within the processing stages, the architecture utilizes a specific intra-stage hybrid strategy. The initial blocks of each stage employ Mamba Mixers to scan the windowed sequences and establish global context, while the subsequent blocks utilize LKA to refine spatial details and texture information. This order ensures that the model first understands the global structure of the leaf and lesion distribution before focusing on the fine-grained visual patterns of the disease. Between stages, depthwise separable convolutions with a stride of 2 are employed for efficient downsampling, followed by additional ECA and GRN modules to preserve feature distinctiveness during resolution reduction.

2.3. MangoMambaBlock

The MangoMambaBlock serves as the foundational computational unit within each stage of the architecture, designed to efficiently extract and refine features for mango leaf disease classification. As illustrated in Figure 7, this block utilizes a dual-pathway residual structure that integrates advanced mixing mechanisms with a robust feed-forward network. The design is engineered to balance the capture of local textural details-essential for identifying specific lesions like Anthracnose-with global context, which is required to detect broader stress symptoms such as Dieback.

To achieve this balance while minimizing computational overhead, the architecture employs an intra-stage hybrid strategy. The initial blocks of a stage utilize the Multi-Scale Mamba Mixer to model long-range dependencies and global sequence patterns with linear complexity. Conversely, the latter blocks of the stage switch to Large-Kernel Attention (LKA) to refine explicit spatial attention and local correlations. This alternation is determined by the block's depth index; empirically, activating LKA in the second half of the stage enhances the model's ability to discriminate between visually similar disease classes by focusing on fine-grained spatial textures after the global context has been established.

The signal propagation within the MangoMambaBlock follows a standard pre-normalization residual design. The input token sequence first undergoes Layer Normalization before entering the designated Mixer module (either Mamba or LKA). The output of the mixer is scaled by a learnable layer-scaling coefficient and subjected to stochastic depth (DropPath) regularization before being added back to the original input. This process is repeated for the second half of the block, which consists of a Layer Normalization step followed by a SwiGLU-based Feed-Forward Network (FFN). The inclusion of learnable layer scaling is critical for stabilizing the training of deep networks by mitigating potential gradient distinctness issues.

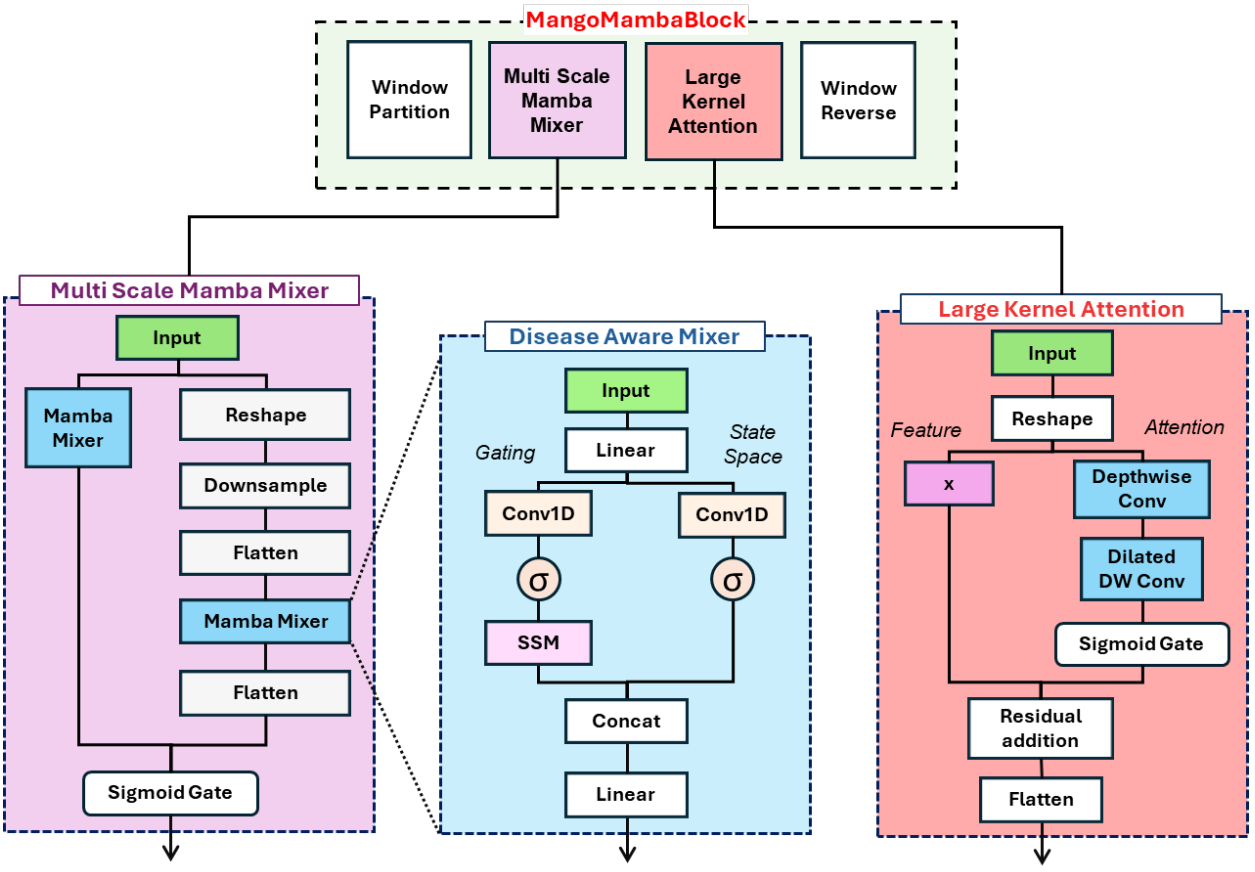


Figure 7: The architectural diagram of the MangoMambaBlock, illustrating the dual-pathway design.

To address the challenge of scale variance in disease symptoms, the Multi-Scale Mamba Mixer processes windowed tokens through two parallel branches. A high-resolution branch processes the native token sequence to preserve fine details, while a low-resolution branch operates on a spatially downsampled version to capture broader semantic features. The low-resolution input is generated via bilinear interpolation, processed, and then upsampled back to the original size. Both branches utilize the Disease Aware Mixer (DAM), and their outputs are fused via a learnable, channel-wise sigmoid gate. This gating mechanism allows the network to adaptively weigh the importance of fine versus coarse features for each channel, optimizing the representation without the parameter explosion typical of dense fusion methods.

The DAM module is a lightweight adaptation of the mixer found in MambaVision [5], specifically optimized for disease pattern recognition. It reduces the state dimension and convolution kernel size to prioritize inference speed on edge devices while preserving the selective reasoning capabilities of State Space Models (SSM). Internally, the input projection is split into two streams. The first stream undergoes a 1D depthwise convolution to capture local context, followed by a selective state-space operation where the discretized state-transition parameters are derived dynamically from the input. This enables the model to selectively propagate or forget information along the sequence, effectively modeling the interconnected nature of disease symptoms across leaf veins. The output is then gated by the second stream (the z-stream) to regulate information flow before the final output projection.

In the later layers of the stage, the block utilizes Large-Kernel Attention to generate spatial attention maps. Unlike the Mamba mixer, LKA is designed to capture spatial correlations directly without the quadratic cost of self-attention. It decomposes a large kernel convolution into a sequence of smaller operations: a standard depthwise convolution to capture local dependencies, followed by a dilated depthwise convolution to expand the receptive field. In this

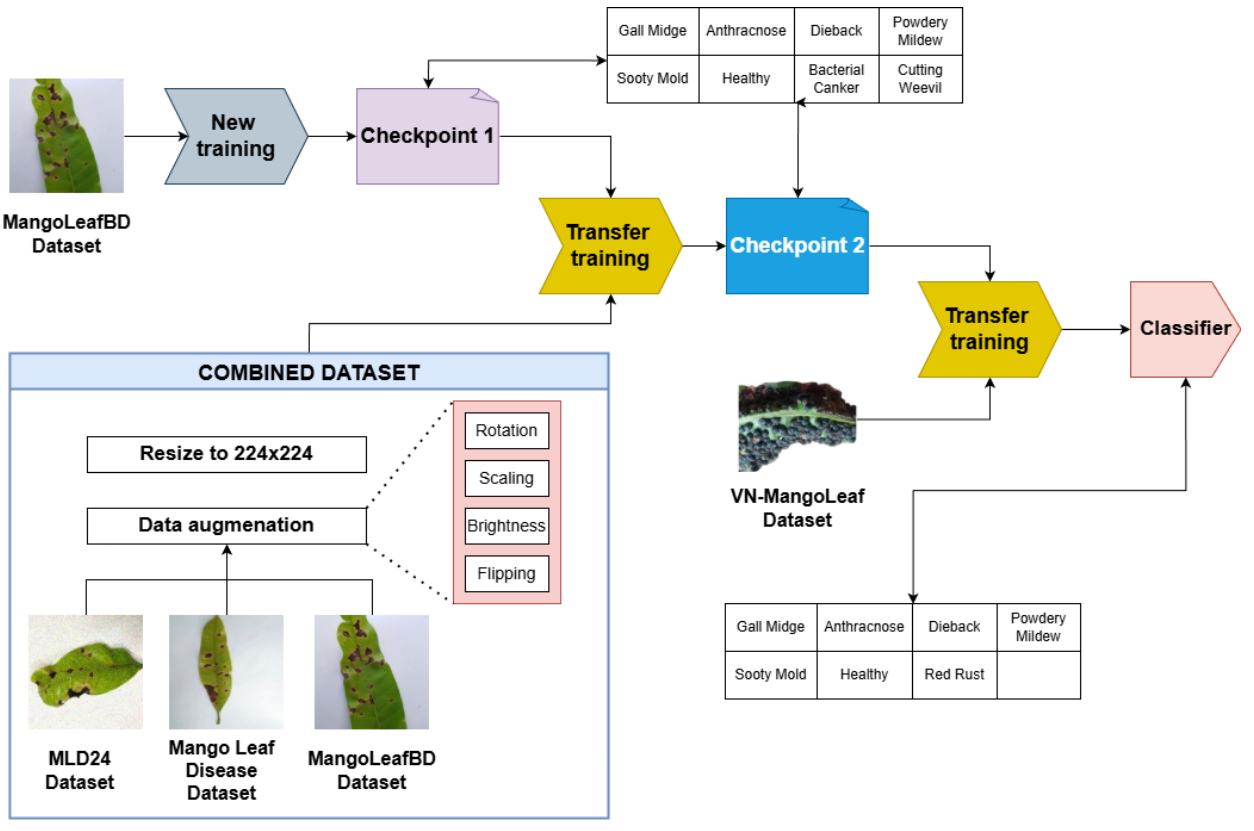


Figure 8: The proposed three-phase training curriculum, progressing from the source dataset (MangoLeafBD) to a combined robust dataset, and finally adapting to the target domain (VN-MangoLeaf).

implementation, a kernel size of 3×3 is used for the local convolution, followed by a 5×5 kernel with a dilation rate of 2. This combination effectively simulates a large receptive field, allowing the model to encompass larger irregular patterns, such as red rust pustules, while remaining computationally efficient. The resulting attention map is applied to the input via element-wise multiplication.

To enhance the non-linear representational capacity of the block, a Position-wise Feed-Forward Network with SwiGLU activation is employed. Selected based on ablation results demonstrating its superiority over standard GELU, the SwiGLU mechanism projects the input into two parallel linear layers. One path is activated by the SiLU function and then element-wise multiplied by the output of the second path. This gating interaction allows the network to learn more complex feature transformations, contributing to the high classification accuracy observed in the experiments.

2.4. Proposed methodology

To address data scarcity and adapt to the specific label-set variations across different agricultural datasets, a three-phase curriculum learning strategy is implemented. This approach, visualized in Figure 8, progressively broadens the data diversity before narrowing down to the specific target domain.

- **Phase 1: Source Domain Initialization (MangoLeafBD).** The MangoMamba backbone is initialized with an 8-class linear projection head. Training is conducted end-to-end on the MangoLeafBD dataset. This phase establishes a foundational representation of canonical mango lesion features, creating a baseline checkpoint.

- **Phase 2: Robust Feature Learning (Combined Dataset).** The checkpoint from Phase 1 is loaded and fine-tuned on a composite dataset constructed from MangoLeafBD, the Mango Leaf Disease Dataset, and MLD24. This phase exposes the model to significant variations in illumination, capture devices, and lesion morphology while maintaining the original 8-way classification scheme. Lower layers are optionally frozen during the initial epochs

to preserve low-level feature extractors, followed by full-model adaptation. This step is critical for learning invariance to environmental conditions.

- **Phase 3: Target Domain Adaptation (VN-MangoLeaf).** In the final phase, the linear classifier is replaced to accommodate the 7-class label set of the target VN-MangoLeaf dataset. The backbone retains weights from the Combined checkpoint. Since the target dataset excludes classes such as 'Cutting Weevil' and 'Bacterial Canker' but emphasizes 'Red Rust', the adaptation process biases the updates toward high-level semantic features while retaining the low-level textural invariances learned in Phase 2. This strategy effectively manages the domain shift and class imbalance inherent in the target environment.

Algorithm 1 formally describes the complete training procedure and the internal forward pass mechanism of the proposed model.

Algorithm 1 MangoMamba Training Curriculum and Inference Procedure

Require: Source Data $\mathcal{D}_{src}$, Combined Data $\mathcal{D}_{comb}$, Target Data $\mathcal{D}_{tgt}$
Ensure: Final Optimized Model $\mathcal{M}_{final}$

Part I: Three-Phase Training Curriculum

$\mathcal{M} \leftarrow \text{InitializeMangoMamba}(classes = 8)$
$\theta_A \leftarrow \text{Train}(\mathcal{M}, \mathcal{D}_{src})$ ▷ Phase 1: Feature Initialization
$\mathcal{M}.\text{load_weights}(\theta_A)$
$\theta_B \leftarrow \text{FineTune}(\mathcal{M}, \mathcal{D}_{comb})$ ▷ Phase 2: Robustness to Variations
$\mathcal{M}.\text{load_weights}(\theta_B)$
$\mathcal{M}.\text{head} \leftarrow \text{Reinitialize}(in = 384, out = 7)$
$\mathcal{M}_{final} \leftarrow \text{FineTune}(\mathcal{M}, \mathcal{D}_{tgt})$ ▷ Phase 3: Domain Adaptation

Part II: Model Inference (Forward Pass)

function FORWARD(x_{rgb})
    $x \leftarrow \text{PatchEmbed}(x_{rgb})$
    $x \leftarrow \text{ECA}(x) + \text{GRN}(x)$
    for $i = 1$ **to** 4 **do**
        $x_{win} \leftarrow \text{WindowPartition}(x, \text{size} = W_i)$
        for $j = 1$ **to** $Depth_i$ **do**
            $x_{norm} \leftarrow \text{LayerNorm}(x_{win})$
            if $j < \lfloor Depth_i/2 \rfloor$ **then**
                $x_{win} \leftarrow x_{win} + \text{MambaMixer}(x_{norm})$ ▷ Global Context
            else
                $x_{win} \leftarrow x_{win} + \text{LKAMixer}(x_{norm})$ ▷ Local Texture
            end if
            $x_{win} \leftarrow x_{win} + \text{SwiGLU_MLP}(\text{LayerNorm}(x_{win}))$
        end for
        $x \leftarrow \text{WindowReverse}(x_{win})$
        $x \leftarrow \text{StageGRN}(x)$
        if $i < 4$ **then**
            $x \leftarrow \text{Downsample}(x)$
        end if
    end for
    $y \leftarrow \text{LinearHead}(\text{GlobalAvgPool}(x))$
    return y
end function

Table 4
Comprehensive ablation analysis of MangoMamba components evaluated on the MangoLeafBD dataset. To ensure statistical reliability and reproducibility, each experimental case was executed across four independent runs using distinct random seeds {42, 123, 2024, 3333}. All models were trained using a consistent hyperparameter configuration: 100 epochs, batch size of 256, initial learning rate of 0.0005 with 5 warmup epochs, and an early stopping patience of 20. The results are reported as Mean ± Standard Deviation. Model complexity is denoted in millions of parameters (M), while Validation and Test metrics (Accuracy and F1 Score) are measured in percentage. The final selected configuration, achieving the optimal trade-off between stability and performance, is highlighted in bold.
Note: (↑) indicates higher is better; (↓) indicates lower is better.

No.	Model Configuration (Ablation Case)	Params (M)↓	Val Accuracy ↑	Val F1 Score ↑	Test Accuracy ↑	Test F1 Score ↑
1	LKA without Mamba, ECA , GRN, SwiGLU	**3.01**	99.09 ± 0.49	99.22 ± 0.40	98.88 ± 0.63	99.19 ± 0.47
2	Mamba without LKA, ECA , GRN, SwiGLU	6.62	99.09 ± 0.21	99.12 ± 0.20	99.25 ± 0.35	99.12 ± 0.32
3	LKA without Mamba	4.41	98.63 ± 0.77	98.78 ± 0.70	98.44 ± 1.45	98.50 ± 1.34
4	Hybrid without ECA , GRN, SwiGLU	4.41	99.09 ± 0.45	99.22 ± 0.36	99.12 ± 0.63	99.19 ± 0.75
5	Hybrid without ECA	5.81	98.59 ± 0.99	98.78 ± 1.01	98.56 ± 1.52	98.75 ± 1.34
6	Hybrid without SwiGLU	4.41	99.25 ± 0.27	99.31 ± 0.16	99.31 ± 0.38	99.44 ± 0.12
7	Hybrid without GRN	5.81	99.56 ± 0.16	99.56 ± 0.16	**99.50 ± 0.20**	99.62 ± 0.32
8	Hybrid with fixed Window [7, 7, 7, 7]	5.81	**99.62 ± 0.10**	**99.62 ± 0.10**	99.44 ± 0.31	**99.69 ± 0.24**
9	Hybrid Baseline (Original) [8, 8, 14, 7]	5.81	98.59 ± 0.91	98.66 ± 0.85	98.56 ± 0.97	98.57 ± 1.02
10	Hybrid on VN-MangoLeaf without Curriculum	5.81	97.22 ± 0.43	97.28 ± 0.39	98.07 ± 0.34	97.63 ± 0.57

3. Results and Discussion

3.1. Ablation study

To validate the effectiveness of the proposed MangoMamba architecture and justify the inclusion of each specific component, a comprehensive ablation study was conducted on the MangoLeafBD dataset. The experiments were performed using consistent hyperparameters across four independent runs for each configuration to ensure statistical reliability. The results, summarizing the impact of the Mixer core, enhancement modules (ECA, GRN, SwiGLU), and windowing strategies, are presented in Table 4.

The core hypothesis of MangoMamba is that combining State Space Models with Large-Kernel Attention offers a balanced trade-off between global context modeling and parameter efficiency compared to pure architectures. As shown in Table 4, the **Pure LKA** model (Case 1) is the most lightweight (3.01M parameters) but lacks the sequence modeling capabilities required for complex disease patterns, resulting in lower stability. Conversely, the **Pure Mamba** model (Case 4) achieves competitive accuracy (99.25%) but incurs a significant parameter penalty (6.62M), which is less ideal for edge deployment. The **Hybrid** architecture without additional features (Case 3) strikes a balance, achieving 99.12% accuracy with 4.41M parameters. This confirms that integrating Mamba Mixers in early layers to capture global dependencies and LKA in later layers for spatial refinement provides an optimal foundation for the model.

The necessity of the integrated components—Efficient Channel Attention (ECA), Global Response Normalization (GRN), and SwiGLU—was evaluated by selectively removing them from the hybrid baseline.

- **Impact of ECA (Case 6):** Removing the ECA module resulted in a significant increase in variance (±1.52%) and a drop in accuracy to 98.56%. This indicates that the channel-wise attention mechanism is critical for stabilizing feature selection and focusing the model on relevant disease symptoms amidst background noise.
- **Impact of GRN (Case 7):** While removing GRN maintained high accuracy (99.50%), the inclusion of GRN in the final configuration (Case 8) contributed to a higher F1-score (99.69% vs. 99.62%). GRN is retained to prevent feature collapse in deep layers and ensure the model prioritizes high-contrast features, which is essential for generalization across diverse datasets.
- **Impact of SwiGLU (Case 5):** Replacing SwiGLU with standard activations resulted in a slight decrease in F1-score compared to the optimal configuration. The SwiGLU activation enhances the non-linear representational capacity of the MLP blocks, allowing the model to capture more intricate decision boundaries between similar disease classes.

Table 5
Hyperparameter configuration for training MangoMamba.

Hyperparameter	Value
Training Epochs	100
Batch Size	256
Base Learning Rate	5×10^{-4}
Warmup Epochs	5
Optimizer	AdamW
Learning Rate Scheduler	Cosine Annealing
Early Stopping Patience	20 epochs

A critical finding from the ablation study is the impact of window size configuration. The initial design utilized a multi-scale window strategy ([8, 8, 14, 7]) to mimic hierarchical receptive fields (Case 9). However, empirical results demonstrate that a **Fixed Window** strategy of [7, 7, 7, 7] (Case 8) yields better performance, achieving the highest F1-score of 99.69% with the lowest standard deviation ($\pm$ 0.24%). This suggests that a consistent local context size of 7×7 aligns better with the typical spatial extent of mango leaf lesions at the processed feature resolutions, minimizing boundary artifacts often introduced by varying window sizes. Consequently, the Fixed Window configuration (Case 8) was selected as the final architecture for MangoMamba.

3.2. Training setup

The experimental validation of the MangoMamba model was conducted within a standardized computational environment to ensure reproducibility and accessibility for resource-constrained agricultural applications. All training and evaluation procedures were executed on the Google Colab platform (Free Tier), utilizing a single NVIDIA Tesla T4 GPU (15 GB VRAM) and 12.7 GB of system RAM. The software stack was built upon Python 3.12 and PyTorch 2.6 with CUDA 12.6 support. Key dependencies included NumPy 2.2.2, Matplotlib 3.10.3, and Scikit-learn 1.7.1 for data handling and analysis; Transformers 4.50, TIMM 1.0.9, and Einops 0.8.1 for model architecture components; and Mamba-SSM 2.2.4 for the state-space modeling operations.

As delineated in Section 2.1, the datasets were stratified into distinct subsets: 80% for training, 10% for validation, and 10% for testing. To ensure assessment of generalization capability, all performance metrics reported in subsequent sections are derived exclusively from the held-out test set, which comprises images completely unseen during the optimization process.

The training configuration was standardized across all experiments to maintain consistency. The specific hyperparameters are detailed in Table 5. The model was optimized using the AdamW optimizer with a cosine annealing learning rate scheduler, initialized at a learning rate of 5×10^{-4} following a 5-epoch linear warmup period. This warmup strategy is critical for stabilizing the early training dynamics of the hybrid Mamba-LKA architecture.

Despite the substantial volume of training data, a comprehensive suite of regularization techniques was employed to mitigate overfitting and simulate the noise characteristics of real-world deployment. Label smoothing was applied to prevent the model from becoming over-confident in its predictions, thereby encouraging softer probability distributions. Weight decay was utilized to penalize large weights and promote parameter sparsity. Stochastic depth (DropPath) was implemented to randomly drop residual paths during training, effectively training an implicit ensemble of sub-networks. Additionally, gradient clipping was enforced to stabilize the optimization of the recurrent state-space components, and an early stopping mechanism with a patience of 20 epochs was used to halt training when validation performance plateaued. These measures collectively ensure that the final deployed model maintains robust performance on edge devices.

3.3. Evaluation metrics

To assess the performance of our proposed model for classifying multiple mango leaf diseases on edge devices, MangoMamba was evaluated on four widely adopted evaluation metrics: Accuracy, Precision, Recall, and F1 Score. These metrics provide a comprehensive evaluation of the model's classification performance across diverse disease classes.

Accuracy quantifies the proportion of correctly classified samples relative to the total number of samples. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

where TP (True Positives) is the number of correctly predicted positive samples, TN (True Negatives) is the number of correctly predicted negative samples, FP (False Positives) is the number of negative samples incorrectly predicted as positive, and FN (False Negatives) is the number of positive samples incorrectly predicted as negative.

Precision measures the proportion of correctly predicted positive samples among all samples predicted as positive, emphasizing the model's ability to minimize false positives. It is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

Recall evaluates the model's ability to identify all positive samples, reflecting its capacity to detect all instances of a specific disease. It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

F1 Score provides a balanced measure of Precision and Recall, beneficial for imbalanced datasets where one class may dominate. It is also the harmonic mean of Precision and Recall, computed as:

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

3.4. Confusion Matrix

To evaluate the classification consistency across different data distributions, the confusion matrices for the MangoLeafBD, Combined, and VN-MangoLeaf test sets are presented in Figure 9 and Figure 10.

Figures 9(a) and 9(b) illustrate the performance on the MangoLeafBD and Combined datasets, respectively, which both encompass the standard eight-class schema (Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Healthy, Powdery Mildew, and Sooty Mold). A distinct dominance of diagonal elements is observed in both matrices, indicating high precision and recall rates. The minimal off-diagonal values suggest that the model successfully extracted discriminative features for each class, maintaining high separability even when the dataset size and diversity were expanded in the Combined set.

In the case of the VN-MangoLeaf dataset (Figure 10), which operates on a seven-class schema (excluding Bacterial Canker and Cutting Weevil while introducing Red Rust), the diagonal structure remains prominent. However, minor misclassifications are observable, particularly between 'Red Rust' and 'Anthracnose'. This phenomenon can be attributed to the morphological similarities between the early-stage necrotic spots of Anthracnose and the reddish-brown pustules of Red Rust, which can pose a challenge for feature extraction under varying lighting conditions. Despite these subtle inter-class similarities, the overall concentration of predictions along the diagonal confirms the reliability of the MangoMamba architecture for practical disease diagnosis.

3.5. Grad-CAM visualization

To validate the interpretability of the model and ensure that predictions rely on pathological features rather than background noise, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed. This technique generates a coarse localization map highlighting the varying intensity of regions within the input image that most significantly influence the decision-making process of the final convolutional layer.

Consistent with the visual analysis framework employed in recent studies [9], the activation heatmaps were overlaid on representative samples from each class to verify semantic alignment. As depicted in Figure 11, the visualizations demonstrate a strong correlation between high-activation regions (red/yellow zones) and the biological symptoms of the diseases:

- For **Powdery Mildew**, the heatmap accurately localizes the white, dusty fungal patches spread across the leaf surface.

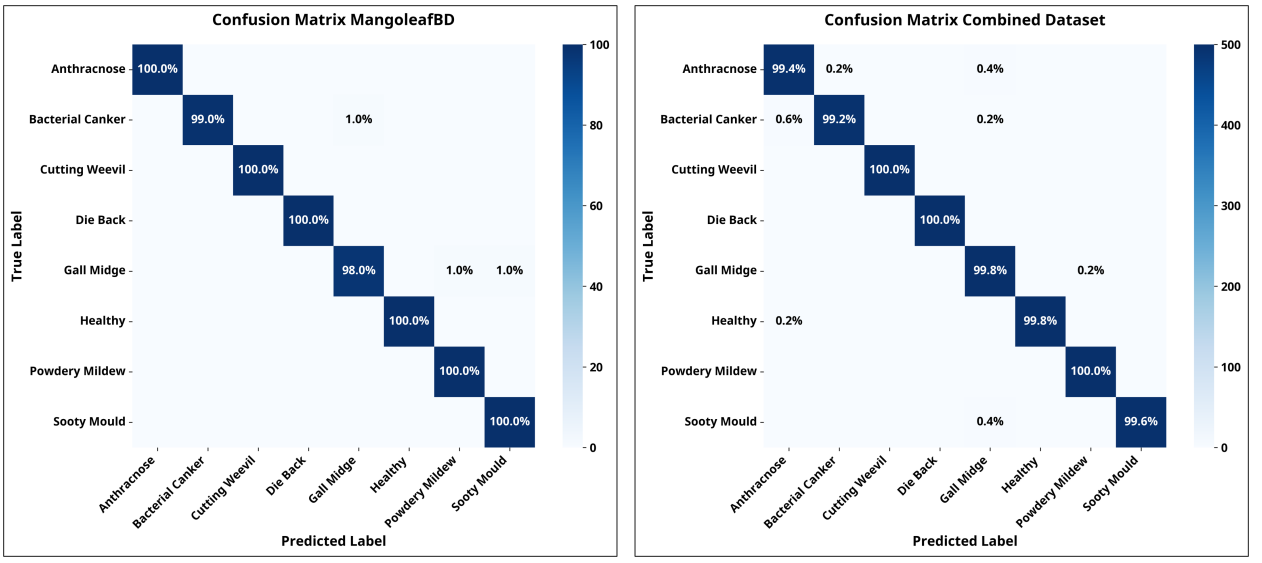


Figure 9: Confusion matrices evaluating classification performance on (a) the MangoLeafBD dataset and (b) the Combined dataset, sharing the same eight-class structure.

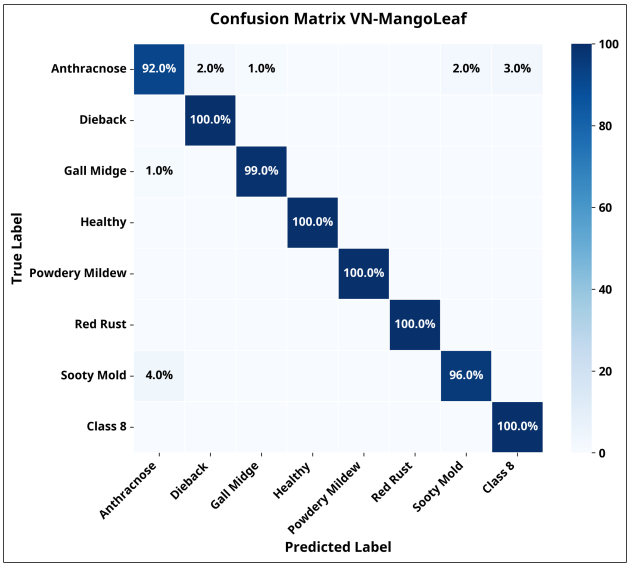


Figure 10: Confusion matrix for the VN-MangoLeaf dataset, illustrating performance across its specific seven-class schema.

- In cases of **Bacterial Canker** and **Anthracnose**, the model focuses intensely on the characteristic necrotic lesions and dark spots, ignoring the healthy green tissues.
- The visualizations for **Sooty Mold** and **Red Rust** confirm the model's attention to the black fungal coatings and reddish-brown pustules, respectively.
- For the **Healthy** class, the activation maps are characteristically diffuse with lower intensity, reflecting the absence of specific lesion patterns.

These qualitative results confirm that the proposed MangoMamba architecture effectively learns distinct disease markers and does not overfit to irrelevant background artifacts.

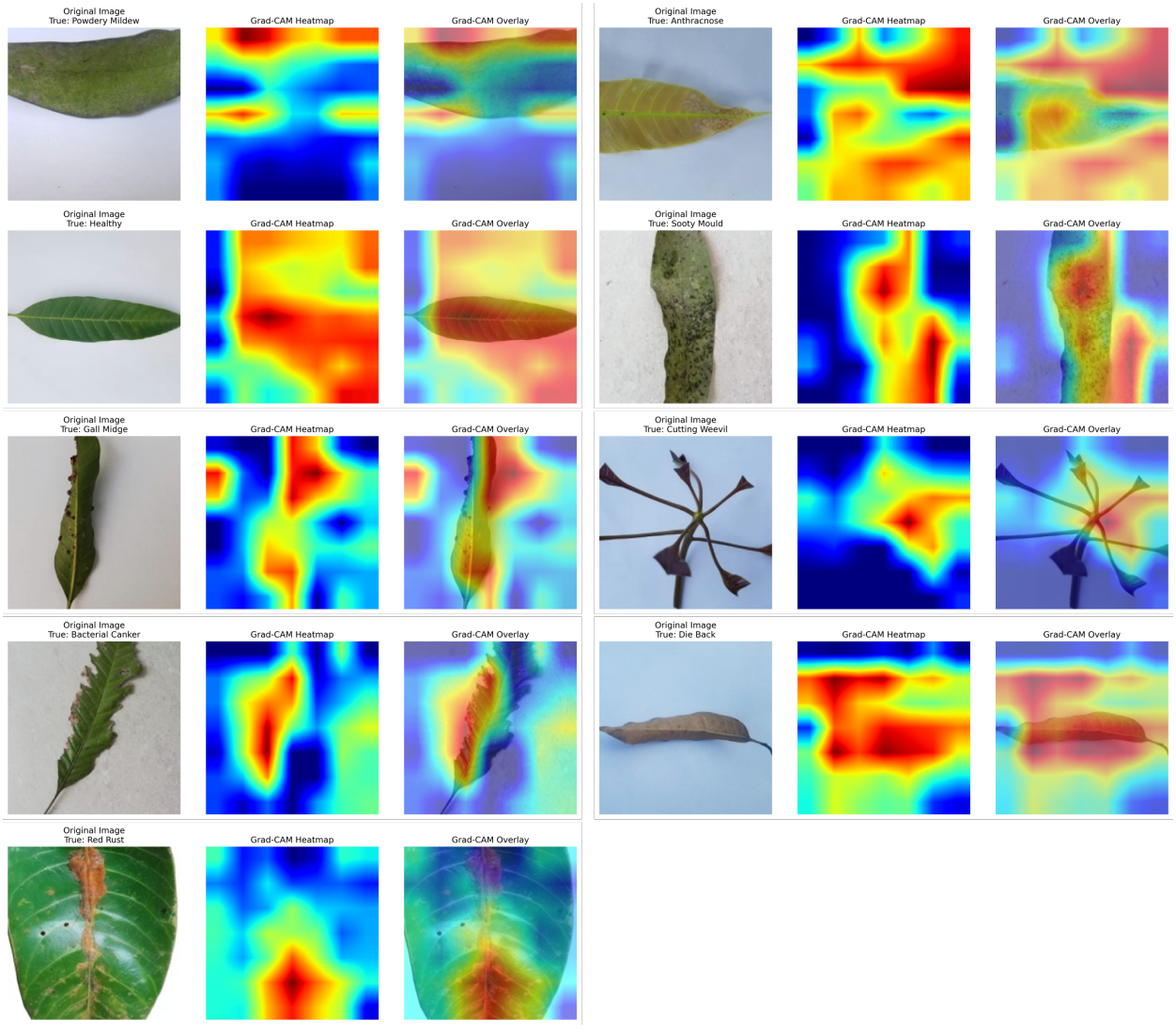


Figure 11: Grad-CAM visualization of representative images from each class. The columns display the original image, the activation heatmap, and the superposition of the heatmap on the original image, demonstrating the model's focus on disease-specific symptoms.

3.6. Performance comparison

Table 6 presents a comprehensive comparison of the proposed MangoMamba model against existing state-of-the-art (SOTA) approaches evaluated on mango leaf disease datasets. The comparison encompasses classification accuracy, computational efficiency (inference time and parameter count), and storage requirements suitable for edge deployment. It is important to note that direct comparisons of inference speed must account for hardware heterogeneity. As the source code and deployed mobile applications for the comparative studies were not publicly available, the performance metrics for these methods are cited directly from their respective publications. Specifically, the models analyzed by Hossain et al. (2024) [citation needed] were evaluated on an Intel Xeon CPU @ 2.20GHz, whereas the ViX-MangoEFormer by Noman et al. (2025) [9] reported inference speeds on mid-tier GPUs (e.g., NVIDIA Jetson Xavier NX or T4 - which is the same type as Google Colab's T4 GPU).

The proposed MangoMamba model demonstrates competitive classification performance, achieving an accuracy of 99.75% on the MangoLeafBD dataset, 99.65% on the Combined dataset, and 98.14% on the newly introduced

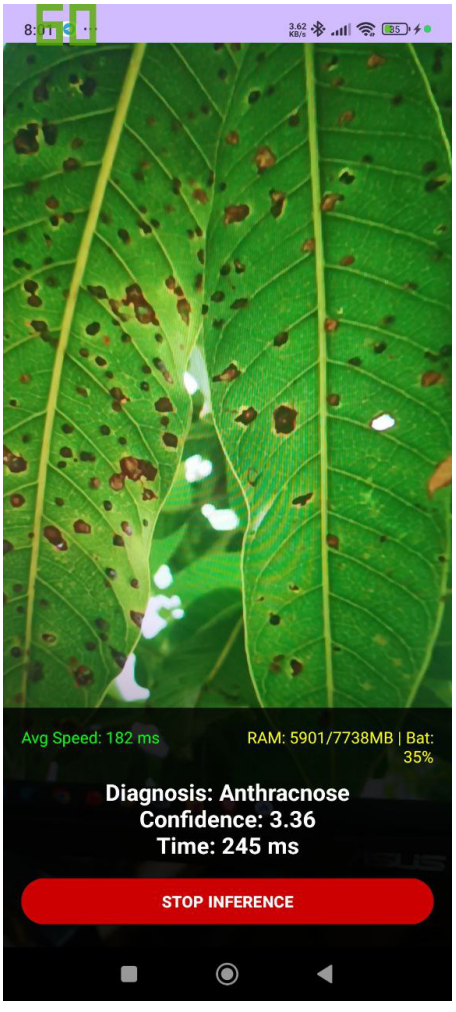


Figure 12: Demonstration of the MangoMamba model deployed on a mobile device via an Android application. The model runs locally using the ONNX runtime, enabling real-time disease classification without internet dependency.

VN-MangoLeaf dataset. While certain heavy architectures such as Fused ViT-ConvNeXt [2] or InceptionV3 [18] may report marginally higher accuracies (e.g., 99.87% on MangoLeafBD), these gains often come at the cost of significantly increased computational complexity, rendering them less suitable for on-device deployment. A substantial advantage of the proposed architecture lies in its inference speed. On a standard GPU environment, MangoMamba requires only 1.46-1.52 ms per image. When compared to the recent ViX-MangoEFormer [9], which reports an inference time of approximately 120 ms on GPU hardware, MangoMamba achieves an approximate 80× reduction in latency while maintaining comparable accuracy.

Unlike many existing studies that primarily utilize models as remote backends accessed via Web APIs (e.g., ViX-MangoEFormer), the proposed framework explicitly supports offline, on-device execution. To validate this capability, the trained MangoMamba model was converted to the ONNX format and integrated into a native Android application using Android Studio.

Figure 12 illustrates the real-world deployment of the application. The system successfully processes images directly on the mobile device, with an inference latency of 100-300 ms (depending on the specific mobile chipset), ensuring functionality even in areas with limited or no internet connectivity. This contrasts with server-dependent solutions that suffer from latency induced by network transmission.

Regarding model size, MangoMamba strikes an optimal balance with 5.8 million parameters and a file size of 22.2 MB. This is significantly more compact than transformer-based alternatives, such as DeiT (85.8M parameters, 1.03

Table 6
Comparison between the proposed model and other studies that also evaluated on the MangoLeafBD dataset. Note: (↑) indicates higher is better; (↓) indicates lower is better.

Model	Accuracy(%) ↑	Dataset	Image Amount ↑	Inference Time (Per image) ↓	Inference Time on mobile ↓	Params ↓	Model Size ↓	Train/Test split
DeiT [6]	**99.75**	MangoLeafBD	4,000	346.87 ms (CPU)	Not provided	85.8 M	1.03 GB	8:1:1
Optimized DeiT [6]	**99.75**	MangoLeafBD	4,000	315.78 ms (CPU)	Not provided	781 K	88.47 MB	8:1:1
SwinTransformer [6]	99.5	MangoLeafBD	4,000	206.02 ms (CPU)	Not provided	27.58 M	110.30 MB	8:1:1
ShuffleNet [6]	99.5	MangoLeafBD	4,000	40.76 ms (CPU)	Not provided	5.36 M	64.72 MB	8:1:1
MobileNetV2 [6]	98.79	MangoLeafBD	4,000	42.28 ms (CPU)	Not provided	2.23 M	27.19 MB	8:1:1
EfficientNet [6]	98.75	MangoLeafBD	4,000	45.01 ms (CPU)	Not provided	4.01 M	48.7 MB	8:1:1
DenseNet121 [6]	96.75	MangoLeafBD	4,000	119.34 ms (CPU)	Not provided	6.96 M	84.47 MB	8:1:1
SqueezeNet [6]	96.5	MangoLeafBD	4,000	20.80 ms (CPU)	Not provided	**726.6 K**	**8.78 MB**	8:1:1
ViX-MangoEFormer [9]	99.43	MangoLeafBD	4,000	120 ms (GPU)	Not provided	Not provided	Not provided	9:1
	98.02	Mango Leaf Disease	12,730	120 ms (GPU)	Not provided	Not provided	Not provided	9:1
	98.63	MLD24	6,400	120 ms (GPU)	Not provided	Not provided	Not provided	9:1
	99.78	Combined	25,530	120 ms (GPU)	Not provided	Not provided	Not provided	9:1
MangoMamba	99.75	MangoLeafBD	4000	**1.46 ms (GPU)**	**100-300 ms**	5.8 M	22.2 & 68.8 MB	8:1:1
Our proposed model	99.72	Combined	**40,000**	1.52 ms (GPU)	**100-300 ms**	5.8 M	22.2 & 68.8 MB	8:1:1
	98.14	VN-MangoLeaf	7000	1.48 ms (GPU)	**100-300 ms**	5.8 M	22.2 & 68.8 MB	8:1:1

GB) and SwinTransformer (27.58M parameters, 110.30 MB). Although SqueezeNet offers a smaller footprint (8.78 MB), its accuracy (96.5%) is notably lower than that of the proposed method (99.75%).

In summary, the experimental results indicate that MangoMamba achieves an effective trade-off between accuracy and efficiency, validating its suitability for real-time edge computing in smart agriculture.

3.7. Error analysis

While the MangoMamba model achieves high overall classification accuracy, a granular analysis of misclassified samples reveals specific limitations in distinguishing phenotypically similar diseases. Figure 13 illustrates representative error cases accompanied by Grad-CAM visualizations, which serve to diagnose whether failures stem from incorrect spatial localization or feature ambiguity.

Two primary misclassification patterns are observed. The first involves confusion between *Anthracnose* and *Red Rust* (Rows 1, 2, and 4 in Figure 13). In these instances, the model incorrectly predicts the class with moderate confidence (e.g., 0.511-0.665). The Grad-CAM heatmaps demonstrate that the attention mechanism correctly activates on the necrotic lesions, indicating that the model successfully localizes the pathology. However, the decision boundary remains ambiguous due to the shared reddish-brown chromatic characteristics of both diseases. This suggests that the feature extractor prioritizes dominant color features over the subtle textural differences (e.g., the powdery pustules of Red Rust versus the sunken necrosis of Anthracnose).

The second pattern involves the misclassification of *Gall Midge* as *Sooty Mold* (Row 3). Both conditions manifest as dark, clustered masses on the leaf surface. Although the model accurately focuses on the gall structures, the morphological similarity in low-contrast regions leads to erroneous predictions.

These findings imply that while the hybrid Mamba-Attention architecture excels at capturing global context and localizing symptoms (as evidenced by precise Grad-CAM activation), it occasionally struggles with fine-grained texture discrimination when inter-class color similarity is high. The relatively low confidence scores in these misclassified cases suggest that integrating an uncertainty threshold mechanism could effectively flag such ambiguous samples for expert review in practical deployment.

3.8. Limitations and Future Research Directions

While the experimental results demonstrate that the proposed MangoMamba architecture achieves a favorable trade-off between computational efficiency and classification accuracy, several limitations regarding the study's scope and deployment validity are acknowledged.

First, regarding the hardware deployment claims, although the model was designed for edge devices, the current empirical validation of real-time inference is limited to a specific mid-range smartphone configuration (Xiaomi Redmi Note 11S). While this device confirms the feasibility of the architecture with an inference latency suitable for interactive use, the performance on lower-end devices or different mobile chipsets (e.g., older Snapdragon series or MediaTek entry-level chips) has not been comprehensively benchmarked. Furthermore, the current deployment relies on the raw PyTorch mobile runtime. The potential gains from hardware-specific optimizations, such as quantization (INT8) or conversion to ONNX/TensorFlow Lite formats, which are critical for broad cross-platform compatibility, remain an open area for investigation.

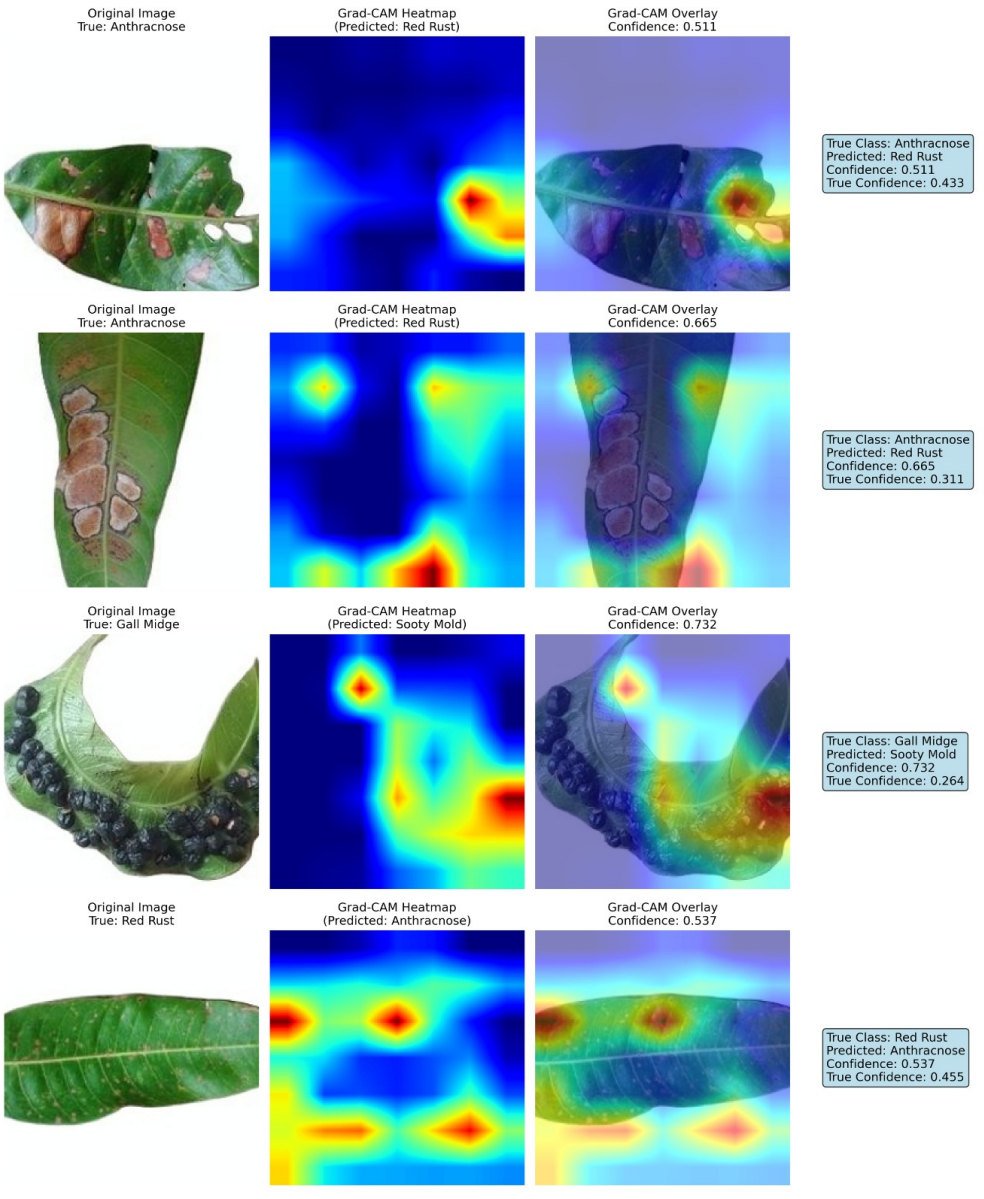


Figure 13: Incorrect predicted images

Second, the generalization capability of the VN-MangoLeaf dataset presents geographical constraints. Although the dataset successfully addresses the bias of existing Bangladesh-centric repositories by introducing Vietnamese-specific disease variants like Red Rust, the data collection was primarily concentrated in Southern Vietnam. Consequently, the model may exhibit reduced sensitivity to disease manifestations specific to Northern Vietnam or other tropical regions with differing climatic conditions. Additionally, the current taxonomy includes seven distinct classes; however, other economically significant mango pathologies, such as Bacterial Black Spot (*Xanthomonas campestris*) or Malformation disease, are absent from the training distribution. This limitation implies that the system classifies unseen pathologies as one of the existing seven categories rather than identifying them as "unknown" or "out-of-distribution."

Third, while data augmentation and curriculum learning were employed to mitigate environmental variability, the robustness of the model under extreme field conditions—such as varying sunlight angles at dusk/dawn, severe occlusion

by overlapping foliage, or motion blur caused by wind—requires further rigorous field testing. The current evaluation relies heavily on static images, whereas real-world agricultural diagnosis often involves dynamic video streams.

Future research directions will focus on three key objectives to address these shortcomings.

- **Cross-Platform ptimization:** Future work will implement and evaluate model quantization and pruning techniques to establish a comprehensive benchmark across a diverse fleet of Android and iOS devices, ensuring consistent real-time performance regardless of hardware tier.
- **Dataset expansion:** The VN-MangoLeaf dataset will be expanded through a citizen science approach, incorporating samples from diverse geographical latitudes to capture broader symptom variations. Concurrently, the architecture will be adapted to support open-set recognition, enabling the model to flag unknown diseases rather than forcing a misclassification.
- **Integrated pest management system:** Beyond singular leaf disease classification, future iterations aim to integrate the MangoMamba backbone into a multi-modal system capable of analyzing fruit quality and pest infestation simultaneously, providing a holistic diagnostic tool for precision agriculture.

4. Conclusion

This study introduced MangoMamba, a lightweight hybrid architecture that effectively balances classification accuracy with computational efficiency for mango leaf disease detection. By integrating Multi-Scale Mamba Mixers with Large-Kernel Attention, the proposed model achieved competitive accuracies of 99.75% on the MangoLeafBD dataset and 98.71% on the newly curated VN-MangoLeaf dataset (7,000 images). With a compact footprint of 5.8 million parameters and a file size of 22.2 MB, the model demonstrates linear complexity, enabling an inference speed of 1.46 ms per image on GPU, representing an approximate 80× speedup compared to recent Transformer-based approaches like ViX-MangoEFormer. Crucially, practical deployment feasibility was validated via an ONNX-based Android application, achieving stable inference latencies of 100-300 ms on mobile devices without internet dependency. Ablation studies confirmed that the synergistic combination of Mamba mixers and attention mechanisms significantly minimizes feature redundancy while maximizing diagnostic precision. Future work will focus on integrating uncertainty quantification to handle ambiguous cases and expanding the dataset to include diverse environmental conditions for broader agro-ecological generalization.

Conflicts of interest

The authors have no conflicts of interest to declare that are relevant to the content of this article.

Data and code availability

To promote transparency and facilitate future research in the domain of agricultural plant pathology, the VN-MangoLeaf dataset has been made publicly available. It is released under the **Creative Commons Attribution 4.0 International (CC BY 4.0)** license, allowing for unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The full dataset, including raw and augmented images, can be accessed via the Kaggle repository at: `https://www.kaggle.com/datasets/binhpt310/viet-nam-mango-diseases-leaves-dataset`.
The source code can also be found at `https://github.com/binhpt310/Mango-Mamba-and-VN-MangoLeaf`.
The corresponding author can provide the datasets and codes generated and/or analyzed during the current work upon reasonable request.

Acknowledgements

This research was supported by The VNUHCM-University of Information Technology's Scientific Research Support Fund.

References

[1] Sarder Iftekhar Ahmed, Muhammad Ibrahim, Md. Nadim, Md. Mizanur Rahman, Maria Mehjabin Shejunti, Taskeed Jabid, and Md. Sawkat Ali. MangoLeafBD: A comprehensive image dataset to classify diseased and healthy mango leaves. *Data in Brief*, 47:108941, April 2023.

[2] Faten S. Alamri, Tariq Sadad, Ahmed S. Almasoud, Raja Atif Aurangzeb, and Amjad Khan. Mango disease detection using fused vision transformer with convnext architecture. *Computers, Materials & Continua*, 83(1):1023–1039, 2025.
[3] Albert Gu and Tri Dao. Mamba: linear-time sequence modeling with selective state spaces, 2023.
[4] Meng-Hao Guo, Cheng-Ze Lu, Zheng-Ning Liu, Ming-Ming Cheng, and Shi-Min Hu. Visual attention network, December 2023.
[5] Ali Hatamizadeh and Jan Kautz. Mambavision: A hybrid mamba-transformer vision backbone. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25261–25270, June 2025.
[6] Md. Arban Hossain, Saadman Sakib, Hasan Muhammad Abdullah, and Shifat E. Arman. Deep learning for mango leaf disease identification: A vision transformer perspective. *Heliyon*, 10(17):e36361, September 2024.
[7] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9992–10002, Montreal, QC, Canada, October 2021. IEEE.
[8] Bahar Uddin Mahmud, Abdullah Al Mamun, Md Jakir Hossen, Guan Yue Hong, and Busrat Jahan. Light-Weight Deep Learning Model for Accelerating the Classification of Mango-Leaf Disease. *Emerging Science Journal*, 8(1):28–42, February 2024.
[9] Abdullah Al Noman, Amira Hossain, Anamul Sakib, Jesika Debnath, Hasib Fardin, Abdullah Al Sakib, Rezaul Haque, Md. Redwan Ahmed, Ahmed Wasif Reza, and M. Ali Akber Dewan. Vix-mangoeformer: an enhanced vision transformer–efficientformer and stacking ensemble approach for mango leaf disease recognition with explainable artificial intelligence. *Computers*, 14(5):171, May 2025.
[10] Ngoh D. J. Patrice, Heu Alain, Mboussi S. Bertrand, Kuate T. W. Norbert, Kone N. A. Nourou, Tchoupou T. D. Brice, Amani G. Honorine, Jeutsa A. Dolaris, Djile Bouba, and Ambang Zachee. First report of red rust disease caused by cephaleuros virescens on mango (Mangifera indica) tree in cameroon. *International Journal of Phytopathology*, 9(3):187–193, December 2020.
[11] Juan Qin, Linfan Deng, Cong Li, Junjie He, Haibo Pen, and Zhaoxia Wang. CropCapsNet: Enhanced capsule network for crop disease classification. *Computers and Electrical Engineering*, 128:110635, December 2025.
[12] Md Shajedur Rahman, Rakibul Hasan, and Mayen Uddin Mojumdar. Mango leaf disease dataset, November 2024.
[13] Yongming Rao, Wenliang Zhao, Benlin Liu, Jiwen Lu, Jie Zhou, and Cho-Jui Hsieh. Dynamicvit: efficient vision transformers with dynamic token sparsification. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*, NIPS '21, Red Hook, NY, USA, 2021. Curran Associates Inc.
[14] Redwan Ahmed Rizvee, Tasnim Hossain Orpa, Adil Ahnaf, Md Ahsan Kabir, Mohammad Rifat Ahmmad Rashid, Mohammad Manzurul Islam, Maheen Islam, Taskeed Jabid, and Md Sawkat Ali. LeafNet: A proficient convolutional neural network for detecting seven prominent mango leaf diseases. *Journal of Agriculture and Food Research*, 14:100787, December 2023.
[15] Md. Mahir Hasan Shakib, Sumaya Mustofa, and Md Taimur Ahad. MLD24: An image dataset for mango leaf disease detection, August 2024.
[16] Noam Shazeer. Glu variants improve transformer, 2020.
[17] Gia-Bao Truong, Thi-Thao Tran, Nhu-Linh Than, Van Quang Nguyen, Thi Hue Nguyen, and Van-Truong Pham. SC-MambaFew: Few-shot learning based on Mamba and selective spatial-channel attention for bearing fault diagnosis. *Computers and Electrical Engineering*, 123:110004, April 2025.
[18] Teena Varma, Prajwal Mate, Noamaan Abdul Azeem, Sanjeev Sharma, and Bhupendra Singh. Automatic mango leaf disease detection using different transfer learning models. *Multimedia Tools and Applications*, May 2024.
[19] Qilong Wang, Banggu Wu, Pengfei Zhu, Peihua Li, Wangmeng Zuo, and Qinghua Hu. Eca-net: efficient channel attention for deep convolutional neural networks. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11531–11539, Seattle, WA, USA, June 2020. IEEE.
[20] Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. Convnext v2: Co-designing and scaling convnets with masked autoencoders, June 2023.
[21] Hewei Zhang, Shengzhou Li, Jialong Xie, Zihan Chen, Jiyang Chen, and Jianwen Guo. VMamba for plant leaf disease identification: design and experiment. *Frontiers in Plant Science*, 16:1515021, April 2025.
[22] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang. Vision mamba: efficient visual representation learning with bidirectional state space model. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org, 2024.

Appendix A: Model Architecture Equations

This appendix provides the detailed mathematical formulations for the core components of the proposed **MangoMamba** architecture, specifically the LayerNorm-SwiGLU-Residual pathway, the Large-Kernel Attention (LKA) mechanism, and the Selective State Space Model (SSM) components.

A.1. Efficient Channel Attention (ECA)

ECA mechanism [19] generates channel-wise attention weights s via a parameter-efficient 1D convolution over global average pooled (GAP) features according to Equation 6:

$$\begin{aligned} s &= \sigma(\text{Conv1D}(\text{GAP}(x))), \\ \hat{x} &= x \odot s, \end{aligned} \tag{6}$$

where sigmoid activation σ is applied on the result of a 1D convolution over the GAP features, and $\hat{x}$ applies these weights to the original input x through element-wise multiplication $\odot$.

A.2. Global Response Normalization (GRN)

GRN mechanism [20] stabilizes feature responses by normalizing the spatial L2-norm (energy) across the height and width dimensions, followed by a learnable affine transformation. This is applied according to Equation 7:

$$\begin{aligned} n_x &= \frac{\|x\|_2}{\left(\text{mean}_C\left(\|x\|_2\right)+\epsilon\right)}, \\ \text{GRN}(x) &= x+\gamma\left(x \odot n_x\right)+\beta, \end{aligned} \tag{7}$$

where $\|\cdot\|_2$ denotes the L 2 -norm computed over the spatial dimensions (H, W) for each channel, $\text{mean}_C(\cdot)$ is the mean across channels, ϵ is a small constant for numerical stability (typically 10^{-6}), $\odot$ represents element-wise multiplication, x and γ are learnable parameters initialized to zero.

A.3. Windowed processing

In each stage, the feature maps are partitioned into non-overlapping windows of size $S \times S$ to facilitate efficient local computation. This process transforms the input tensor $x \in \mathbb{R}^{B\times C\times H\times W}$ into a windowed representation $X \in \mathbb{R}^{B_{\text{win}}\times L\times C}$ ($L = S^2$ where L represents the number of tokens within each window) and according to Equation 8:

$$B_{\text{win}} = B \cdot \left(\frac{H}{s}\right) \cdot \left(\frac{W}{s}\right), \tag{8}$$

where B_{win} is the total number of windows, with B being the batch size, while $\frac{H}{s}$ and $\frac{W}{s}$ indicate the number of windows along the height and width dimensions, respectively.

A.4. Dual-pathway residual architecture

Structurally, the MangoMambaBlock employs a dual-pathway residual architecture, comprising a LayerNorm-Mixer-Residual pathway followed by a LayerNorm-SwiGLU-Residual pathway. For an input token sequence $x \in \mathbb{R}^{B\times L\times C}$ the forward pass is formalized as Equation 9:

$$\begin{aligned} x' &= x+\text{DropPath}\left(\gamma_1 \cdot \text{Mixer}(\text{ LayerNorm }(x))\right), \\ y &= x'+\text{DropPath}\left(\gamma_2 \cdot \text{FFN}\left(\text{LayerNorm}\left(x'\right)\right)\right), \end{aligned} \tag{9}$$

where Mixer denotes either the Multi-Scale Mamba Mixer or LKA based on the block position, FFN is the SwiGLU-based feed-forward network, DropPath applies stochastic depth regularization with a stage-specific drop rate (derived from a linear schedule up to 0.15), and γ_1, γ_2 are learnable layer scaling coefficients initialized to 1×10^{-5} to mitigate vanishing gradients and promote stable training, as per modern practices in vision transformers [13].

A.5. Multi-Scale Mamba Mixer

The Multi-Scale Mamba Mixer addresses the challenge of capturing dependencies at varying spatial resolutions by processing windowed tokens through parallel high- and low-resolution branches. The high-resolution branch processes the full token sequence of length $L = S^2$, while the low-resolution branch operates on a spatially downsampled version of length $\frac{L}{4}$, achieved by reshaping to $\mathbb{R}^{B\times C\times S\times S}$, applying bilinear interpolation downsampling to $\frac{S}{2}\times\frac{S}{2}$, and flattening. Each branch utilizes a DiseaseAwareMixer (DAM) module, with the high branch employing an expansion factor of 1.5 for richer representations and the low branch using 1.0 for efficiency. The outputs of $y_{\text{high}} = \text{DAM}(x)$ and $y_{\text{low}} = \text{Upsample}(\text{DAM}(\text{Downsample}(x)))$ are fused using a learnable per-channel gate as Equation 10:

$$\begin{aligned} &g = \sigma(W) \in \mathbb{R}^{1\times 1\times c}, \\ &y = (1-g)y_{\text{high}} + g\,y_{\text{low}} \end{aligned} \tag{10}$$

where W is a learnable parameter tensor initialized to zero (shape $[1, 1, C]$) and σ is the sigmoid activation. This gating mechanism allows adaptive blending, reducing parameters compared to dense fusion while enhancing multi-scale feature integration.

A.6. DiseaseAwareMixer

The input x is first projected to an inner dimension $d_{\text{inner}} = \lfloor \text{expand} \times C \rfloor$ where $z = \text{in_proj}(x)$. This expanded representation is split into x-stream x_z (first half) and z-stream z_z (second half) for state-space processing and gating, respectively. The x-stream undergoes SiLU activation and depthwise 1D convolution for local context according to Equation 11:

$$\begin{aligned} x_{\text{conv}} &= \text{SiLU}\left(\text{DWConv1D}\left(x_z, \text{kernel} = 3\right)\right), \\ \left(\Delta_t, B_t, C_t\right) &= x_\text{proj}\left(x_{\text{conv}}\right), \end{aligned} \tag{11}$$

where Δ_t is further projected and softplus-activated via dt_proj to ensure positive time steps. The selective state-space recurrence is then computed using the selective_scan_fn interface [from mamba_ssm] as follow in Equation 12:

$$\begin{aligned} h_t &= \tilde{A}\left(\Delta_t\right) h_{t-1} + \tilde{B}\left(\Delta_t, B_t\right) x_t \\ o &= h_t C_t \\ y &= o \odot \text{SiLU}\left(z_z\right) \end{aligned} \tag{12}$$

with $\tilde{A}$ and $\tilde{B}$ as discretized matrices, enabling input-dependent evolution of the hidden state $h_t \in \mathbb{R}^{d_{\text{state}}}$ then the output is gated by the SiLU-activated z-stream and projected back to C dimensions via out_proj. This formulation provides efficient modeling of long-range dependencies in windowed tokens, which are crucial for detecting interconnected disease symptoms across leaf veins.

A.7. Large-Kernel Attention (LKA)

In later blocks, the Large-Kernel Attention (LKA) pathway replaces the Mamba Mixer to generate spatial attention maps through cascaded depthwise convolutions, achieving large receptive fields without quadratic complexity. This is advantageous for capturing multi-scale spatial dependencies in mango leaves, such as the irregular patterns of red rust pustules. The input tokens are reshaped to spatial format $x \in \mathbb{R}^{B \times C \times S \times S}$, then processed sequentially. A depthwise convolution with kernel size $k_1 = 3$, padding 1 followed by a dilated depthwise convolution with $k_2 = 5$, dilation 2 will result a residual gate with a learnable per-channel parameter $G \in \mathbb{R}^{1 \times C \times 1 \times 1}$ as described in Equation 13 :

$$\begin{aligned} a &= \text{DWConv}_{k_1=3}(x), \\ b &= \text{DWConv}_{k_2=5, d=2}(a), \\ y &= x + \sigma(G) \odot b, \end{aligned} \tag{13}$$

Where σ is sigmoid and $\odot$ is broadcasted element-wise multiplication. The output is flattened back to token sequence format for downstream processing.

A.8. SwiGLU activation

To further enhance representational capacity, each block incorporates a position-wise feed-forward network using SwiGLU activation. The input is projected to a reduced hidden dimension $h = \lfloor 0.8\times \text{ mlp_ratio} \times C \rfloor$ (mlp_ratio=2.0 typically) via two parallel linear layers V_1 and V_2 followed by gating Equation 14:

$$\begin{aligned} \text{SiLU}(z) &= z \cdot \sigma(z), \\ \text{SwiGLU}(x) &= \left(\text{SiLU}\left(x_{V2}\right) \odot x_{V1}\right) W_o, \end{aligned} \tag{14}$$

where $\odot$ is element-wise, and W_o is the output projection back to C.